

# Systematic Review of Pharmacologic Treatment for Migraine Prevention in Adults

Report of the AAN Guidelines Subcommittee and the American Headache Society

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## Abstract

### Background and Objectives

This systematic review (SR) provides updated evidence-based conclusions regarding the use of pharmacologic migraine prevention in adults to inform a new joint American Academy of Neurology (AAN) and American Headache Society practice guideline.

### Methods

A multidisciplinary panel conducted an SR following the 2017 AAN Clinical Practice Guideline Process Manual. Randomized controlled trials evaluating pharmacologic preventive treatments for adults with episodic or chronic migraine were included. Searches encompassed MEDLINE, Embase, and ClinicalTrials.gov from database inception through June 6, 2024. Studies were screened in duplicate, with dual independent risk-of-bias assessment. Outcomes included change in monthly headache days,  $\geq 50\%$  responder rate, and validated patient-reported quality of life (QOL) measures. Raw mean differences, standardized mean differences, and risk ratios were calculated. A modified Grading of Recommendations Assessment, Development, and Evaluation process was used to classify certainty of evidence.

### Results

A total of 217 studies met inclusion criteria. For episodic migraine, high-confidence evidence showed that galcanezumab and erenumab are more effective than placebo in reducing headache frequency. Moderate-confidence evidence supported benefit from atogepant, eptinezumab, fremanezumab, propranolol, topiramate, and valproate. Several additional oral agents including amitriptyline, bisoprolol, flunarizine, fluoxetine, levetiracetam, metoprolol, nifedipine, pizotifen, and telmisartan had low-confidence evidence suggesting possible benefit. For chronic migraine, high-confidence evidence supported reductions in headache frequency with fremanezumab, galcanezumab, and onabotulinumtoxinA. Moderate-confidence evidence supported benefit from atogepant, eptinezumab, erenumab, topiramate and valproate. Across both episodic and chronic migraine populations, erenumab, fremanezumab, galcanezumab, eptinezumab, rimegepant, atogepant, topiramate and onabotulinumtoxinA demonstrated improvements in patient-reported QOL outcomes on validated instruments. Evidence comparing active treatments was limited and generally of low or very low confidence, restricting conclusions about comparative effectiveness.

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### Supplementary Material

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## Glossary

AAN = American Academy of Neurology; AHS = American Headache Society; CGRP = calcitonin gene-related peptide; COI = conflicts of interest; CV = curriculum vitae; FDA = U.S. Food and Drug Administration; GRADE = Grading of Recommendations Assessment, Development and Evaluation; GS = Guidelines Subcommittee; HIT-6 = Headache Impact Test; MIDAS = Migraine Disability Assessment Scale; MPFID-EA = Migraine Physical Function Impact Diary-Everyday Activity; MPFID-PI = Migraine Physical Function Impact Diary-Physical Impairment; MSQ-EF = Migraine Specific Quality of Life Questionnaire-Emotional Function; MSQ-RFP = Migraine Specific Quality of Life Questionnaire-Role Function Preventive; MSQ-RFR = Migraine Specific Quality of Life Questionnaire-Role Function Restriction; NMDA-R = NMDA receptor; PACAP = pituitary adenylate cyclase-activating polypeptide; QOL = quality of life; RCT = randomized controlled trial; RMD = raw mean difference; ROB = risk of bias; RR = risk ratio; SC = subcutaneous; SMD = standardized mean difference; SR = systematic review.

## Discussion

This SR provides a comprehensive synthesis of evidence on pharmacologic migraine prevention in adults. High- and moderate-confidence findings confirm the efficacy of several established and newer preventive therapies and demonstrate improvements in patient-reported outcomes across multiple validated measures. These conclusions informed the development of evidence-based recommendations, presented in a companion publication, to guide clinicians in selecting preventive medications for adults with episodic and chronic migraine.

## Introduction

On any given day, 15.8% of the world's population may be experiencing headache.<sup>1</sup> Worldwide, migraine is the second leading cause of years lived with disability.<sup>2</sup> In the United States, headache disorders are the third leading cause of disability adjusted life years in women, and the fifth in men, most of which are migraine.<sup>3</sup> Economic modeling using US data from 2007 estimated that migraine resulted in \$11 billion in direct costs<sup>4</sup> and indirect costs of \$12 billion that year.<sup>5</sup>

Preventive medications are recommended for migraine based on frequency and disability.<sup>6</sup> These medications reduce overall headache burden<sup>7</sup> and improve patient-reported quality of life (QOL).<sup>8</sup> Yet, only one-third of people with migraine who should be considered for these treatments use them.<sup>9</sup>

The previous American Academy of Neurology (AAN) and American Headache Society (AHS) joint practice guideline examining pharmacologic migraine prevention in adults was published in 2012.<sup>7</sup> Since then, new studies have examined the effectiveness of medications for both episodic and chronic migraine prevention, including a new class of treatments targeting calcitonin gene-related peptide (CGRP).

The purpose of this review is to systematically examine all high-quality randomized controlled trials (RCTs) that evaluate the effectiveness of pharmacologic preventive treatments for episodic and chronic migraine and to provide clinicians with recommendations regarding their use. The term chronic migraine refers to headache occurring

≥15 days/month for >3 months, which, on at least 8 days/month, has the features of migraine headache (taking acute migraine-specific medication can be defined as a migraine day). This systematic review focuses on the following clinical questions:

1. In adults (aged 18 years and older) with migraine, which pharmacologic preventive treatments, compared with placebo, reduce headache frequency (change in number of headache days per month, 50% responder rate)?
2. In adults (aged 18 years and older) with migraine, which pharmacologic preventive treatments, compared with other active preventive pharmacologic treatments, reduce headache frequency (change in number of headache days per month, 50% responder rate)?
3. In adults (aged 18 years and older) with migraine, which pharmacologic preventive treatments, compared with placebo or other preventive pharmacologic treatments, improve patient-reported QOL on validated instruments?

## Analytic Process

In January 2018, the Guidelines Subcommittee (GS) of the (AAN) (eAppendices 1 and 2) convened a panel of experts to develop this guideline. In March 2022, the AHS joined the project as a collaborating organization. The author panel includes content experts (W.J.B., R.C.B., L.C., J.F., A.H., J.L.J., E.L., T.S., S.S.), methodology experts (T.P., D.S.), current and former AAN GS members (S.P., M.M., M.O., S.T., G.L.G.P., V.K.R.), an AHS representative (A.H.), 2 patient representatives and advocates (S.K., M.M.), and AAN staff (T.F., K.B.D., H.S.). Each potential

author submitted an AAN relationship disclosure form and their curriculum vitae (CV). The panel leadership, consisting of the lead developer (S.P.), methodologist (T.P.), and GS leadership, reviewed the disclosure forms and CVs for financial and intellectual conflicts of interest (COI). These documents were screened to exclude individuals with a clear financial conflict and those whose professional and intellectual bias would diminish the credibility of the review. As required by the AAN, at least 51 percent of the panel members and the lead developer (S.P.) are free of COI relevant to the guideline. Six of the 19 developers were determined to have relevant COI, which were judged to be not significant enough to preclude them from authorship (W.J.B., A.H., S.K., E.L., T.S., S.S.). These developers were not permitted to review or rate the evidence and were consulted in an advisory capacity to help with the validation of the key questions, the scope of the literature search, and the identification of seminal articles to validate the literature search. Because the panel majority is free of COI, the entire panel voted on the recommendations, with members free of COI leading the recommendation development process. This author panel was solely responsible for decisions concerning the design, analysis, and reporting of the proposed systematic review (SR) and subsequent guideline.

The SR included randomized clinical trials and study results published on ClinicalTrials.gov on pharmacologic treatments for migraine in adults with episodic or chronic migraine, compared with placebo or another active pharmacologic treatment. The panel identified 2 SRs that synthesized studies of oral preventive treatments published up to May 2014,<sup>10</sup> and studies of onabotulinumtoxinA injections published up to March 2018.<sup>11</sup> The guideline developers agreed to use the data from these SRs in the evidence synthesis process and update the results. The panel searched the MEDLINE, Embase, and ClinicalTrials.gov databases from database inception to November 14, 2018, for relevant peer-reviewed articles that met inclusion criteria (Figure; eAppendix 3 for search strategies). The initial search yielded 3,340 articles. AAN staff conducted a preliminary review to remove nonhuman studies and screen for relevance and appropriate study type. The panelists reviewed 1,688 article titles and abstracts for potential relevance. Of the reviewed abstracts, 103 were identified as potentially relevant and their corresponding articles were obtained for full-text review. Selected articles were RCTs with acceptable study designs that contained information relevant to the clinical questions. Reviews, meta-analyses, case-control studies, cohort studies, case series, studies including participants who had unrelated diseases or were outside of the study population, and articles that were not peer reviewed were excluded. Of the 103 articles reviewed, 44 were rated for risk of bias (ROB) by 2 panel members working independently of each other following the process in the 2017 AAN process manual, as amended.<sup>12</sup> Differences between the 2 raters were reconciled by raters re-

viewing with input from their counterpart. In addition, articles reviewed and rated from the 2 SRs<sup>10,11</sup> were incorporated, adding 178 and 107 articles respectively. The panelists ultimately selected 162 articles for evidence rating.

An updated literature search was conducted in June 2024. The search yielded 12,666 articles, many of which were not germane to the search due to the Embase search strategy not being appropriately narrowed and were removed by AAN staff. The panelists reviewed 2,260 article titles and abstracts for potential relevance. Of those, 434 were considered potentially relevant and moved on to full-text review. One hundred twenty-eight articles were then moved to ROB assessment. After 2 reviewers assessed each article for ROB, 55 articles were selected for evidence rating. These were combined with those previously selected, bringing the total number of studies included in this SR to 217.

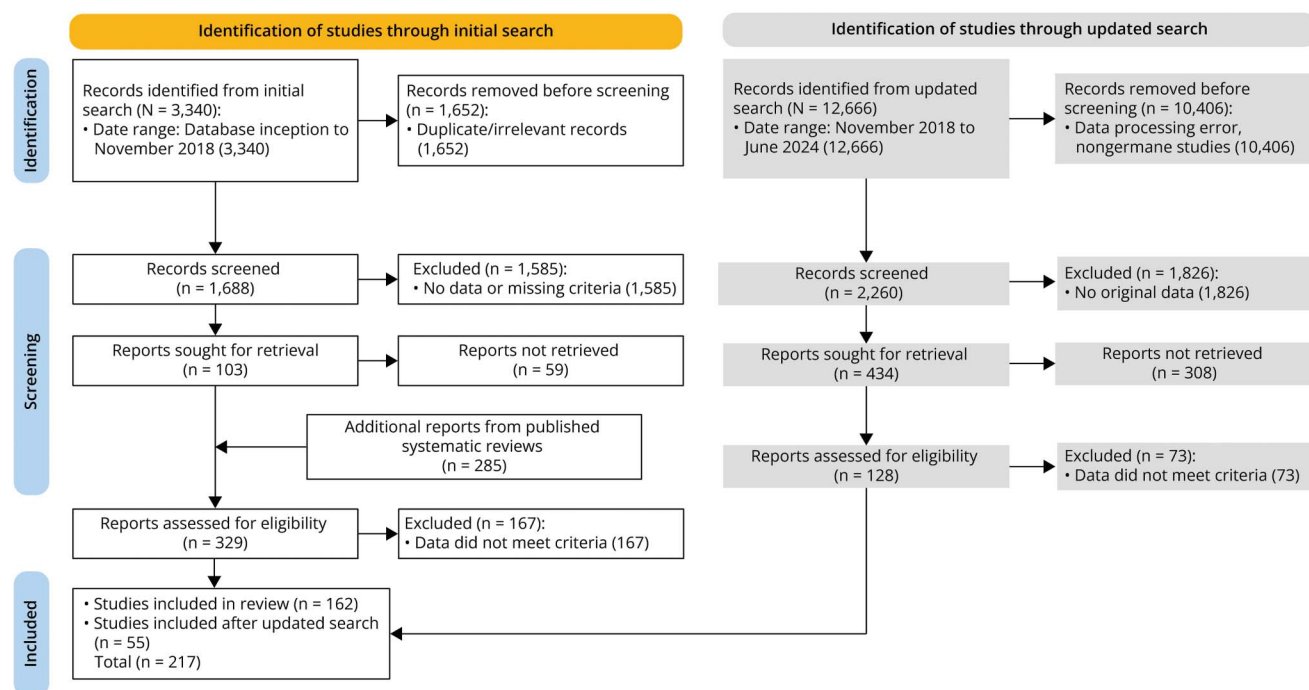
The following outcome measures were included:

1. Change in migraine frequency through assessment of:
  - a. Change in number of days with migraine or headache (if migraine days were not reported) per month, using the raw mean difference (RMD) in comparisons with placebo, and the standardized mean difference (SMD) in comparisons with active treatments, based on reporting from the SR of Jackson et al.<sup>10</sup>
  - b. Responder rate, defined as the proportion of participants with a  $\geq 50\%$  decrease in migraine frequency compared with baseline, using the risk ratio (RR).
2. Change in patient-reported QOL outcomes on validated scales using the RMD, including:
  - a. Migraine Disability Assessment Scale (MIDAS).<sup>13,14</sup>
  - b. Headache Impact Test (HIT-6).<sup>15</sup>
  - c. Migraine Specific Quality of Life Instrument–Role Function Restriction (MSQ-RFR), Role Function Preventive (MSQ-RFP), and Emotional Function (MSQ-EF) domains.<sup>16,17</sup>
  - d. Migraine Physical Function Impact Diary–Everyday Activities (MPFID-EAs) and Physical Impairment (MPFID-PI) domains.<sup>18</sup>

For the purposes of making recommendations, for all included drugs, we consulted the Lexicomp database and US Food and Drug Administration (FDA) drug labels for evidence-based information about adverse reactions, warnings and precautions, and drug effects on reproduction, pregnancy, and lactation.

A modified form of the Grading of Recommendations Assessment, Development, and Evaluation process was used to develop conclusions. In this process, the evidence is analyzed based on various parameters of ROB, consistency, directness, precision, and publication bias. This process permits

**Figure** Study Selection Flow Chart



transparency in the upgrading or downgrading of evidence classification.

The panel used the evidence-based conclusions and stipulated axiomatic principles of care to develop rationales and recommendation to inform clinical practice. This process, and the resulting recommendations, are reported in a companion publication.<sup>19</sup>

## Standard Protocol Approvals, Registrations, and Patient Consents

The protocol for this guideline was posted for a 30-day public comment period in July 2018; the comments received were reviewed by the author panel, and changes were incorporated into the manuscript, as appropriate. This project was not required to be reviewed by an institutional review board, and this SR has not been registered.

## Data Availability

Any data not published within this article are available from the AAN, by request.

## Analysis of Evidence

Table 1 presents a simplified summary of the confidence in the evidence for preventive medications for episodic and chronic migraine vs placebo for change in monthly headache days at 12 weeks (or closest available time point).

## Clinical Question 1

1. In adults (aged 18 years and older) with migraine, which pharmacologic preventive treatments, compared with placebo, reduce headache frequency (change in number of headache days per month, 50% responder rate)?

## Episodic Migraine

For each drug, Table 2 presents the dose, time point, effect size, and evidence to support each conclusion about reducing headache frequency when used for episodic migraine prevention. Throughout this article, we have indicated drug dosages that are currently unavailable clinically or that differ from the recommended dosages on the drug label.

In addition to the data in Table 2, based on moderate confidence, oral topiramate 50 mg/day is probably no more likely than placebo to reduce headache frequency at 16 weeks (2 Class II studies<sup>43,44</sup>). Based on low confidence, the following drugs are possibly no more likely than placebo to reduce headache frequency: eptinezumab 100 mg IV every 12 weeks at 48 weeks (1 Class II study<sup>56</sup>), oral fluoxetine 40 mg/day at 8 weeks (1 Class II study<sup>59</sup>), oral gabapentin 1,200 mg/day at 4 weeks (1 Class II study<sup>e8</sup>), oral nimodipine 120 mg/day at 4 weeks (1 Class II and 1 Class III study<sup>e4,e9</sup>), onabotulinumtoxinA 7.5–50 units IM injection at 12 weeks (1 Class II study<sup>e10</sup>), oral oxcarbazepine 1,200 mg/day at 15 weeks (1 Class II study<sup>e11</sup>), and oral topiramate 50 mg/day at 20 and 24 weeks (2 Class II studies<sup>43,44</sup>). eAppendix 4 lists drugs for which there was insufficient evidence



**Table 1** Clinician's Overall Summary Table

|                                                                                                                      | Episodic migraine headache days                                                                                                                                                                                                        | Chronic migraine headache days                                                |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| High confidence in the evidence                                                                                      | Erenumab<br>Galcanezumab                                                                                                                                                                                                               | Fremanezumab<br>Galcanezumab<br>OnabotulinumtoxinA                            |
| Moderate confidence in the evidence                                                                                  | Atogepant<br>Eptinezumab<br>Fremanezumab<br>Propranolol<br>Rimegepant<br>Topiramate<br>Valproate                                                                                                                                       | Atogepant<br>Eptinezumab<br>Erenumab<br>Topiramate<br>Rimegepant<br>Valproate |
| Low confidence in the evidence                                                                                       | Amitriptyline<br>Bisoprolol<br>Flunarizine<br>Fluoxetine<br>Levetiracetam<br>Metoprolol<br>Nifedipine<br>Pizotifen<br>Telmisartan                                                                                                      |                                                                               |
| Very low confidence in the evidence (insufficient evidence to determine if treatment is more effective than placebo) | Atenolol<br>Candesartan<br>Clomipramine<br>Clonazepam<br>Clonidine<br>Duloxetine<br>Enalapril<br>Gabapentin<br>Lamotrigine<br>Lisinopril<br>Nicardipine<br>Oxprenolol<br>Pindolol<br>Timolol<br>Venlafaxine<br>Verapamil<br>Vigabatrin | Atenolol<br>Propranolol<br>Tizanidine                                         |

(very low confidence) to determine superiority to placebo for reducing headache frequency.

For each drug, Table 3 presents the dose, time point, effect size, and evidence to support each conclusion about the responder rate when used for episodic migraine prevention.

In addition to the data in Table 3, based on low confidence, oral flunarizine 5–10 mg/day is possibly no more effective than placebo in reducing headache frequency by  $\geq 50\%$  at 16 weeks (1 Class II study<sup>e16</sup>). eAppendix 4 lists drugs for which there was insufficient evidence (very low confidence) to determine superiority to placebo for reducing headache frequency by  $\geq 50\%$ .

### Chronic Migraine

For each drug, Table 4 presents the dose, time point, effect size, and evidence to support each conclusion about reducing headache frequency when used for chronic migraine prevention.

For each drug, Table 5 presents the dose, time point, effect size, and evidence to support each conclusion about responder rate when used for chronic migraine prevention.

Based on low confidence, tizanidine is possibly no more effective than placebo in reducing headache frequency at 12 weeks (1 Class II study).<sup>e30</sup> eAppendix 4 lists drugs for which there was insufficient evidence (very low confidence) to determine superiority to placebo for reducing headache frequency.

### Episodic or Chronic Migraine

For each drug, eTable 1 presents the dose, time point, effect size, and evidence to support each conclusion about headache frequency when used for episodic or chronic migraine prevention.

For each drug, eTable 2 presents the dose, time point, effect size, and evidence to support each conclusion about responder rate when used for episodic or chronic migraine prevention.

### Clinical Question 2

1. In adults (aged 18 years and older) with migraine, which pharmacologic preventive treatments, compared with other active preventive pharmacologic treatments, reduce headache frequency (change in number of headache days per month, 50% responder rate)?

Most evidence comparing active treatments for both episodic and chronic migraine is of low or very low quality. For most comparisons, the evidence only suggests that one drug is no more likely than its comparator to decrease headache frequency, or that there is insufficient evidence to determine if one treatment is superior to another.

### Episodic Migraine: Change in Number of Headache Days per Month

#### Moderate Confidence in the Evidence

Amitriptyline (oral, 25–100 mg/day) is probably no more likely than topiramate (oral, 25–100 mg/day) to reduce headache frequency at 12–26 weeks (SMD 0.006, 95% CI –0.311 to 0.323; 2 Class II studies<sup>e31,e32</sup>).

Galcanezumab (240 mg loading dose followed by 120 mg monthly subcutaneous [SC] injection) is probably no more likely than rimegepant (75 mg oral every other day) to reduce headache frequency at 12 weeks (SMD –0.17, 95% CI –0.33 to 0; 1 Class I study<sup>e33</sup>).

Propranolol (oral, 80–160 mg/day) is probably no more likely than topiramate (oral, 50–200 mg/day) to reduce headache frequency at 12–26 weeks (SMD 0.02, 95% CI –0.30 to 0.34; 2 Class II studies<sup>39,e34</sup>).

**Table 2** Episodic Migraine: Change in Number of Headache Days

| High confidence in the evidence (more likely than placebo to reduce headache frequency)              |                                                             |                                |                                           |                                                   |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------|-------------------------------------------|---------------------------------------------------|
| Drug                                                                                                 | Dose, frequency, and route                                  | Time point                     | RMD (95% CI)                              | Evidence                                          |
| Galcanezumab                                                                                         | 240 mg loading dose followed by monthly 120 mg SC injection | 12 wk                          | −2.13 (−2.88 to −1.38)                    | 2 class I studies <sup>20,21</sup>                |
|                                                                                                      | 240 mg loading dose followed by monthly 120 mg SC injection | 24 wk                          | −2.24 (−2.84 to −1.64)                    | 3 class I studies <sup>22–24</sup>                |
|                                                                                                      | 240 mg monthly SC injection <sup>a</sup>                    | 24 wk                          | −2.08 (−2.59 to −1.57)                    | 3 class I studies <sup>22–24</sup>                |
| Erenumab                                                                                             | 70 mg monthly SC injection                                  | 24 wk                          | −1.75 (−2.34 to −1.17)                    | 3 class I studies <sup>25–27</sup>                |
|                                                                                                      | 140 mg monthly SC injection                                 | 12 wk                          | −1.67 (−2.31 to −1.02)                    | 2 class I studies <sup>28,29</sup>                |
|                                                                                                      | 140 mg monthly SC injection                                 | 24 wk                          | −1.9 (−2.27 to −1.52)                     | 2 class I studies <sup>25,26</sup>                |
| Moderate confidence in the evidence (probably more likely than placebo to reduce headache frequency) |                                                             |                                |                                           |                                                   |
| Drug                                                                                                 | Dose, frequency, and route                                  | Time point                     | RMD (95% CI)                              | Evidence                                          |
| Atogepant                                                                                            | 10 mg oral once daily                                       | 12 wk                          | −1.2 (−1.67 to −0.73)                     | 2 class I studies <sup>30,31</sup>                |
|                                                                                                      | 30 mg oral once daily                                       | 12 wk                          | −1.18 (−1.67 to −0.7)                     | 2 class I studies <sup>30,31</sup>                |
|                                                                                                      | 30 mg oral twice daily <sup>a</sup>                         | 12 wk                          | −1.4 (−2.2 to −0.6)                       | 1 class I study <sup>30</sup>                     |
|                                                                                                      | 60 mg oral once daily                                       | 12 wk                          | −1.57 (−2.47 to −0.67)                    | 3 class I studies <sup>30–32</sup>                |
|                                                                                                      | 60 mg oral twice daily <sup>a</sup>                         | 12 wk                          | −1.30 (−2.1 to −0.5)                      | 1 class I study <sup>30</sup>                     |
| Eptinezumab                                                                                          | 100 mg IV every 12 wk                                       | 12 wk                          | −0.69 (−1.25 to −0.12)                    | 1 class I study <sup>33</sup>                     |
|                                                                                                      | 300 mg IV every 12 wk                                       | 12 wk                          | −1.11 (−1.68 to −0.54)                    | 1 class I study <sup>33</sup>                     |
| Erenumab                                                                                             | 70 mg monthly SC injection                                  | 12 wk                          | −1.03 (−1.50 to −0.55)                    | 2 class I studies <sup>34,35</sup>                |
| Fremanezumab                                                                                         | 225 mg monthly SC injection                                 | 12 wk                          | −2.23 (−3.7 to −0.76)                     | 2 class I studies <sup>36,37</sup>                |
|                                                                                                      | 675 mg SC injection every 12 wk                             | 12 wk                          | −2.13 (−3.79 to −0.46)                    | 2 class I studies <sup>36,37</sup>                |
| Galcanezumab                                                                                         | 150 mg SC injection every 2 wk <sup>a</sup>                 | 12 wk                          | −1.20 (−1.90 to −0.60)                    | 1 class I study <sup>38</sup>                     |
| Propranolol                                                                                          | 160 mg oral per d                                           | 4 wk                           | −1.35 (−1.84 to −0.86)                    | 2 class II studies <sup>39,40</sup>               |
|                                                                                                      | 60–180 mg oral per d                                        | 8 wk                           | −0.69 (−1.22 to −0.16)                    | 2 class II studies <sup>39,41</sup>               |
|                                                                                                      | 60–180 mg oral per d                                        | 12 wk                          | −1.55 (−2.98 to −0.12)                    | 4 class II studies <sup>39–42</sup>               |
|                                                                                                      | 160–180 mg oral per d                                       | 16 wk                          | −1.01 (−1.82 to −0.20)                    | 2 class II studies <sup>39,41</sup>               |
|                                                                                                      | 60–180 mg oral per d                                        | 20 wk                          | −1.02 (−1.65 to −0.38)                    | 2 class II studies <sup>39,41</sup>               |
| Topiramate                                                                                           | 50 mg oral per d                                            | 4 wk                           | −1.04 (−1.95 to −0.14)                    | 3 class II studies <sup>43–45</sup>               |
|                                                                                                      |                                                             | 8 wk                           | −0.67 (−1.13 to −0.22)                    | 2 class II studies <sup>42,43</sup>               |
|                                                                                                      | 100 mg oral per d                                           | 4 wk                           | −0.90 (−1.32 to −0.48)                    | 3 class II studies <sup>39,43,44</sup>            |
|                                                                                                      |                                                             | 8 wk                           | −0.95 (−1.35 to −0.55)                    | 3 class II studies <sup>39,43,44</sup>            |
|                                                                                                      |                                                             | 12 wk                          | −0.87 (−1.35 to −0.39)                    | 3 class II studies <sup>39,43,44</sup>            |
|                                                                                                      |                                                             | 16 wk                          | −0.91 (−1.36 to −0.45)                    | 3 class II studies <sup>39,43,44</sup>            |
|                                                                                                      |                                                             | 20 wk                          | −0.94 (−1.42 to −0.45)                    | 3 class II studies <sup>39,43,44</sup>            |
|                                                                                                      |                                                             | 24 wk                          | −1.23 (−1.72 to −0.74)                    | 3 class II studies <sup>43,44,46</sup>            |
|                                                                                                      |                                                             | 200 mg oral per d <sup>a</sup> | 4 wk                                      | −1.64 (−2.11 to −1.16)                            |
|                                                                                                      | 8 wk                                                        | −1.47 (−2.06 to −0.88)         | 3 class II studies <sup>43,44,47</sup>    |                                                   |
|                                                                                                      | 12 wk                                                       | −1.52 (−1.95 to −1.09)         | 3 class II studies <sup>43,44,47</sup>    |                                                   |
|                                                                                                      | 16 wk                                                       | −1.09 (−1.73 to −0.44)         | 4 class II studies <sup>43,44,47,48</sup> |                                                   |
|                                                                                                      | 20 wk                                                       | −1.38 (−1.82 to −0.95)         | 3 class II studies <sup>43,44,47</sup>    |                                                   |
|                                                                                                      | 24 wk                                                       | −1.44 (−2.00 to −0.88)         | 2 class II studies <sup>43,44</sup>       |                                                   |
| Valproate                                                                                            | 500–1,500 mg oral per d                                     | 12 wk                          | −1.74 (−2.86 to −0.63)                    | 2 class I and 3 class II studies <sup>49–53</sup> |

Continued

**Table 2** Episodic Migraine: Change in Number of Headache Days (*continued*)

Low confidence in the evidence (possibly more likely than placebo to reduce headache frequency)

| Drug          | Dose, frequency, and route | Time point | RMD (95% CI)           | Evidence                                          |
|---------------|----------------------------|------------|------------------------|---------------------------------------------------|
| Amitriptyline | 25 mg oral per d           | 12 wk      | −1.10 (−1.50 to −0.70) | 1 class II study <sup>54</sup>                    |
| Bisoprolol    | 5–10 mg oral per d         | 8 wk       | −0.61 (−1.10 to −0.16) | 1 class II study <sup>55</sup>                    |
| Eptinezumab   | 300 mg IV every 12 wk      | 48 wk      | −1.20 (−1.95 to −0.46) | 1 class II study <sup>56</sup>                    |
| Flunarizine   | 10 mg oral per d           | 16 wk      | −1.20 (−2.00 to −0.41) | 2 class III studies <sup>57,58</sup>              |
| Fluoxetine    | 40 mg oral per d           | 4 wk       | −0.32 (−1.36 to −0.06) | 1 class II study <sup>59</sup>                    |
|               |                            | 12 wk      | −0.74 (−1.44 to −0.03) | 1 class II study <sup>59</sup>                    |
| Levetiracetam | 1,000 mg oral per d        | 8–12 wk    | −2.20 (−3.00 to −1.30) | 1 class II study <sup>60</sup>                    |
| Metoprolol    | 100–200 mg oral per d      | 4–8 wk     | −1.16 (−1.84 to −0.47) | 2 class III studies <sup>e1,e2</sup>              |
| Nifedipine    | 60 mg oral per d           | 4 wk       | −3.80 (−4.80 to −2.80) | 1 class III study <sup>e3</sup>                   |
|               | 120 mg oral per d          | 8 wk       | −1.50 (−2.40 to −0.73) | 1 class II study <sup>e4</sup>                    |
|               |                            | 12 wk      | −1.30 (−1.90 to −0.69) | 1 class II study <sup>e4</sup>                    |
| Pizotifen     | 1.5 mg oral per d          | 4–12 wk    | −0.46 (−0.76 to −0.17) | 1 class II and 1 class III study <sup>e5,e6</sup> |
| Telmisartan   | 80 mg oral per d           | 12 wk      | −1.90 (−3.60 to −0.23) | 1 class II study <sup>e7</sup>                    |
| Topiramate    | 50 mg oral per d           | 12 wk      | −0.55 (−1.04 to −0.05) | 2 class II studies <sup>e3,44</sup>               |

Abbreviations: RMD = raw mean difference; SC = subcutaneous.

<sup>a</sup> Drug dosage is currently not available clinically or differs from the recommended dosages on the drug label.

*Low Confidence in the Evidence*

Atorvastatin (oral, 20 mg/day) plus sodium valproate (oral, 500 mg/day) is possibly more likely than sodium valproate (oral, 500 mg/day) to reduce headache frequency at 8 weeks (SMD −2.33, 95% CI −2.97 to −1.7; 1 Class II study<sup>e35</sup>).

Metoprolol (oral, 200 mg/day) is possibly more likely than flunarizine (oral, 10 mg/day) to reduce headache frequency at 20 weeks (SMD −0.35, 95% CI −0.69 to −0.02; 1 Class II study<sup>e36</sup>).

Pizotifen (oral, 2–3 mg/day) is possibly no more likely than flunarizine (oral, 10 mg/day) to reduce headache frequency at 8–16 weeks (SMD 0.23, 95% CI −0.17 to 0.62; 2 Class II studies<sup>e37,e38</sup>).

Pizotifen (oral, 1.5 mg/day) is possibly no more likely than naproxen (oral, 550 mg/day) to reduce headache frequency at 12 weeks (SMD 0.10, 95% CI −0.27 to 0.46; 1 Class II study<sup>e5</sup>).

Propranolol (oral, 160 mg/day) is possibly more likely than femoxetine (oral, 400 mg/day) to reduce headache frequency at 16 weeks (SMD −2.03, 95% CI −2.66 to −1.39; 1 Class III study<sup>e39</sup>).

Propranolol (oral, 60–180 mg/day) is possibly no more likely than flunarizine (oral, 10 mg/day) to reduce headache frequency at 16–24 weeks (SMD −0.03, 95% CI −0.47 to 0.41; 4 Class II studies<sup>e16,e40–e42</sup> and 1 Class III study<sup>e43</sup>).

Propranolol (oral, 80–160 mg/day) is possibly no more likely than nadolol (oral, 80–160 mg/day) to reduce headache frequency at 8–12 weeks (SMD 0.30, 95% CI −0.03 to 0.62; 2 Class III studies<sup>e44,e45</sup>).

Propranolol (oral, 40 mg/day) is possibly more likely than nortriptyline (oral, 20 mg/day) to reduce headache frequency at 8 weeks (SMD −0.83, 95% CI −1.61 to −0.06; 1 Class II study<sup>e46</sup>).

Topiramate (oral, 100 mg/day) is possibly no more likely than flunarizine (oral, 5 mg/day) to reduce headache frequency at 48 weeks (SMD 0.23, 95% CI −0.07 to 0.53; 1 Class II study<sup>e47</sup>).

Topiramate (oral, 100 mg/day) plus nortriptyline (oral, 30 mg/day) is possibly more likely than topiramate (100 mg) to reduce headache frequency at 10 weeks (SMD −0.53, 95% CI −1.02 to −0.04; 1 Class II study<sup>e48</sup>).

Topiramate (oral, 50 mg/day) is possibly no more likely than valproate (oral, 400 mg/day) to reduce headache frequency at 8–12 weeks (SMD −0.25, 95% CI −0.61 to 0.11; 2 Class III studies<sup>e49,e50</sup>).

Topiramate (oral, 100 mg/day) is possibly no more likely than zonisamide (oral, 200 mg/day) to reduce headache frequency at 12 weeks (SMD −0.26, 95% CI −0.72 to 0.19; 1 Class II study<sup>e51</sup>).

**Table 3** Episodic Migraine: Responder Rate<sup>a</sup>**High confidence in the evidence (more likely than placebo to reduce headache frequency by  $\geq 50\%$ )**

| Drug             | Dose, frequency, and route  | Time point | RR (95% CI)      | Evidence                           |
|------------------|-----------------------------|------------|------------------|------------------------------------|
| <b>Atogepant</b> | 60 mg oral once daily       | 12 wk      | 1.93 (1.25–2.97) | 3 class I studies <sup>30–32</sup> |
| <b>Erenumab</b>  | 70 mg monthly SC injection  | 24 wk      | 2.11 (1.36–3.26) | 3 class I studies <sup>25–27</sup> |
|                  | 140 mg monthly SC injection | 24 wk      | 2.46 (1.28–4.70) | 2 class I studies <sup>25,26</sup> |

**Moderate confidence in the evidence (probably more likely than placebo to reduce headache frequency by  $\geq 50\%$ )**

| Drug                | Dose, frequency, and route                                  | Time point | RR (95% CI)      | Evidence                                              |
|---------------------|-------------------------------------------------------------|------------|------------------|-------------------------------------------------------|
| <b>Atogepant</b>    | 10 mg oral once daily                                       | 12 wk      | 1.65 (1.15–2.15) | 2 class I studies <sup>30,31</sup>                    |
|                     | 60 mg oral twice daily <sup>b</sup>                         | 12 wk      | 1.53 (1.2–1.94)  | 1 class I study <sup>30</sup>                         |
| <b>Eptinezumab</b>  | 300 mg IV every 12 wk                                       | 12 wk      | 1.51 (1.23–1.85) | 1 class I study <sup>33</sup>                         |
| <b>Erenumab</b>     | 70 mg monthly SC injection                                  | 12 wk      | 1.30 (1.16–1.47) | 3 class I studies <sup>29,34,35</sup>                 |
|                     | 140 mg monthly SC injection                                 | 12 wk      | 1.78 (1.22–2.60) | 2 class I and 1 class II studies <sup>28,29,e12</sup> |
| <b>Fremanezumab</b> | 225 mg monthly SC injection                                 | 12 wk      | 2.41 (1.14–5.10) | 2 class I studies <sup>36,37</sup>                    |
|                     | 675 mg SC injection every 12 wk                             | 12 wk      | 1.59 (1.27–1.99) | 1 class I study <sup>36</sup>                         |
| <b>Galcanezumab</b> | 240 mg loading dose followed by monthly 120 mg SC injection | 12 wk      | 1.63 (1.17–2.26) | 3 class I studies <sup>20–22</sup>                    |
|                     | 240 mg loading dose followed by monthly 120 mg SC injection | 24 wk      | 1.48 (1.21–1.82) | 3 class I studies <sup>22–24</sup>                    |
|                     | 240 mg monthly SC injection <sup>b</sup>                    | 24 wk      | 1.44 (1.20–1.72) | 3 class I studies <sup>22–24</sup>                    |
| <b>Propranolol</b>  | 120–160 mg oral per d                                       | 24–26 wk   | 1.72 (1.21–2.42) | 2 class II studies <sup>39,e13</sup>                  |
| <b>Topiramate</b>   | 50–200 mg oral per d                                        | 26 wk      | 1.67 (1.34–2.08) | 2 class II studies <sup>39,43</sup>                   |
| <b>Valproate</b>    | 500–1,500 mg oral per d                                     | 12–26 wk   | 2.04 (1.41–2.94) | 2 class I and 3 class II studies <sup>49–53</sup>     |

**Low confidence in the evidence (possibly more likely than placebo to reduce headache frequency by  $\geq 50\%$ )**

| Drug               | Dose, frequency, and route          | Time point | RR (95% CI)       | Evidence                                |
|--------------------|-------------------------------------|------------|-------------------|-----------------------------------------|
| <b>Atogepant</b>   | 30 mg oral once daily               | 12 wk      | 1.65 (1.06–2.55)  | 2 class I studies <sup>30,31</sup>      |
|                    | 30 mg oral twice daily <sup>b</sup> | 12 wk      | 1.44 (1.1–1.84)   | 1 class I study <sup>30</sup>           |
| <b>Enalapril</b>   | 10 mg oral per d                    | 8 wk       | 5.60 (1.40–21.90) | 1 class II study <sup>e14</sup>         |
| <b>Eptinezumab</b> | 100 mg IV every 12 wk               | 12 wk      | 1.33 (1.07–1.65)  | 1 class I study <sup>33</sup>           |
|                    | 100 mg IV every 12 wk               | 48 wk      | 1.28 (1.17–1.41)  | 1 class II study <sup>56</sup>          |
|                    | 300 mg IV every 12 wk               | 48 wk      | 1.29 (1.17–1.41)  | 1 class II study <sup>56</sup>          |
| <b>Propranolol</b> | 160 mg oral per d                   | 12 wk      | 2.00 (1.20–2.80)  | 1 class II study <sup>42</sup>          |
| <b>Timolol</b>     | 20 mg oral per d                    | 8–12 wk    | 1.90 (1.40–2.50)  | 1 class II study <sup>42</sup>          |
| <b>Topiramate</b>  | 50–200 mg oral per d                | 12–16 wk   | 1.52 (1.04–2.22)  | 3 class II studies <sup>44,47,e15</sup> |

Abbreviation: RR = risk ratio; SC = subcutaneous.

<sup>a</sup> Responder rate = A reduction in headache frequency of  $\geq 50\%$ .<sup>b</sup> Drug dosage is currently not available clinically or differs from the recommended dosages on the drug label.

Valproate (oral, 200–600 mg/day) is possibly no more likely than zonisamide (oral, 50–150 mg/day) to reduce headache frequency at 12 weeks (SMD 0.22, 95% CI –0.20 to 0.64; 1 Class II study<sup>e52</sup>).

Valproate (oral, 400 mg/day) is possibly no more likely than pregabalin (oral, 100 mg/day) to reduce headache frequency at 12 weeks (SMD –0.11, 95% CI –0.53 to 0.31; 1 Class II study<sup>e53</sup>).



**Table 4** Chronic Migraine: Change in Number of Headache Days per Month**High confidence in the evidence (more likely than placebo to reduce headache frequency)**

| Drug                      | Dose, frequency, and route                                                                                                                          | Time point | RMD (95% CI)            | Evidence                                  |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------|-------------------------------------------|
| <b>Fremanezumab</b>       | 675 mg SC injection every 12 wk                                                                                                                     | 12 wk      | −1.77 (−2.26 to −1.28)  | 2 class I studies <sup>e17,e18</sup>      |
| <b>Fremanezumab</b>       | 675 mg at baseline, 275 mg at 4 wk, 275 mg at 8 wk by SC injection                                                                                  | 12 wk      | −1.96 (−2.43 to −1.49)  | 3 class I studies <sup>e17,e18,e19</sup>  |
| <b>Galcanezumab</b>       | 240 mg loading dose followed by monthly 120 mg SC injection                                                                                         | 12 wk      | −2.77 (−4.32 to −1.22)  | 2 class I studies <sup>e21,e20</sup>      |
| <b>OnabotulinumtoxinA</b> | 155–195 units IM injections across corrugator, procerus, frontalis, temporalis, occipitalis, cervical paraspinal, and trapezius muscles every 12 wk | 24 wk      | −1.96 (−2.84 to −1.07)  | 2 class I studies <sup>e21,e22</sup>      |
| <b>Topiramate</b>         | 50–200 mg oral per d                                                                                                                                | 4 wk       | −5.42 (−7.61 to −3.22)  | 3 class II studies <sup>e45,e23,e24</sup> |
|                           | 50–200 mg oral per d                                                                                                                                | 16 wk      | −8.93 (−13.01 to −4.84) | 2 class II studies <sup>e23,e24</sup>     |

**Moderate confidence in the evidence (probably more likely than placebo to reduce headache frequency)**

| Drug                | Dose, frequency, and route                  | Time point | RMD (95% CI)            | Evidence                                  |
|---------------------|---------------------------------------------|------------|-------------------------|-------------------------------------------|
| <b>Atogepant</b>    | 30 mg oral twice daily <sup>a</sup>         | 12 wk      | −2.4 (−3.5 to −1.3)     | 1 class I study <sup>e25</sup>            |
| <b>Atogepant</b>    | 60 mg oral once daily                       | 12 wk      | −1.8 (−2.9 to −0.8)     | 1 class I study <sup>e25</sup>            |
| <b>Eptinezumab</b>  | 100 mg IV every 12 wk                       | 12 wk      | −2.03 (−2.88 to −1.18)  | 1 class I study <sup>e26</sup>            |
| <b>Eptinezumab</b>  | 100 mg IV every 12 wk                       | 24 wk      | −1.98 (−2.94 to −1.01)  | 1 class I study <sup>e26</sup>            |
| <b>Eptinezumab</b>  | 300 mg IV every 12 wk                       | 12 wk      | −2.6 (−3.45 to −1.74)   | 1 class I study <sup>e26</sup>            |
| <b>Eptinezumab</b>  | 300 mg IV every 12 wk                       | 24 wk      | −2.65 (−3.62 to −1.68)  | 1 class I study <sup>e26</sup>            |
| <b>Erenumab</b>     | 70 mg SC injection every 4 wk               | 12 wk      | −2.50 (−3.50 to −1.40)  | 1 class I study <sup>e27</sup>            |
|                     | 140 mg SC injection every 4 wk              | 12 wk      | −2.50 (−3.50 to −1.40)  | 1 class I study <sup>e27</sup>            |
| <b>Fremanezumab</b> | 900 mg SC injection every 4 wk <sup>a</sup> | 12 wk      | −2.74 (−4.60 to −0.90)  | 1 class I study <sup>e19</sup>            |
| <b>Galcanezumab</b> | 240 mg SC injection monthly <sup>a</sup>    | 12 wk      | −1.90 (−2.70 to −1.10)  | 1 class I study <sup>e20</sup>            |
| <b>Topiramate</b>   | 50–200 mg oral per d                        | 8 wk       | −9.14 (−16.30 to −1.98) | 3 class II studies <sup>e45,e23,e24</sup> |
|                     |                                             | 12 wk      | −6.0 (−8.8 to −3.2)     | 1 class II study <sup>e23</sup>           |
| <b>Valproate</b>    | 800 mg oral per d                           | 12 wk      | −4.30 (−7.10 to −1.50)  | 1 class I study <sup>e49</sup>            |

**Low confidence in the evidence (possibly more likely than placebo to reduce headache frequency)**

| Drug             | Dose, frequency, and route | Time point | RMD (95% CI)             | Evidence                        |
|------------------|----------------------------|------------|--------------------------|---------------------------------|
| <b>Valproate</b> | 1,000 mg oral per d        | 4 wk       | −12.60 (−17.90 to −7.30) | 1 class IV study <sup>e28</sup> |

Abbreviation: RMD = raw mean difference; SC = subcutaneous.

<sup>a</sup> Drug dosage is currently not available clinically or differs from the recommended dosages on the drug label.

**Table 5** Chronic Migraine: Responder Rate<sup>a</sup>

| High confidence in the evidence (more likely than placebo to reduce headache frequency by ≥ 50%)              |                                                                                                                                                      |            |                  |                                      |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------|--------------------------------------|
| Drug                                                                                                          | Dose, frequency, and route                                                                                                                           | Time point | RR (95% CI)      | Evidence                             |
| Fremanezumab                                                                                                  | 675 mg SC injection every 12 wk                                                                                                                      | 12 wk      | 2.11 (1.70–2.62) | 2 class I studies <sup>e17,e18</sup> |
|                                                                                                               | 675 mg at baseline, 275 at 4 wk, 275 mg at 8 wk by SC injection                                                                                      | 12 wk      | 2.25 (1.82–2.78) | 2 class I studies <sup>e17,e18</sup> |
| Moderate confidence in the evidence (probably more likely than placebo to reduce headache frequency by ≥ 50%) |                                                                                                                                                      |            |                  |                                      |
| Drug                                                                                                          | Dose, frequency, and route                                                                                                                           | Time point | RR (95% CI)      | Evidence                             |
| Atogepant                                                                                                     | 30 mg oral twice daily <sup>b</sup>                                                                                                                  | 12 wk      | 1.64 (1.28–2.12) | 1 class I study <sup>e25</sup>       |
| Atogepant                                                                                                     | 60 mg oral once daily                                                                                                                                | 12 wk      | 1.58 (1.22–2.04) | 1 class I study <sup>e25</sup>       |
| Eptinezumab                                                                                                   | 100 mg IV every 12 wk                                                                                                                                | 12 wk      | 1.46 (1.25–1.71) | 1 class I study <sup>e26</sup>       |
| Eptinezumab                                                                                                   | 100 mg IV every 12 wk                                                                                                                                | 24 wk      | 1.39 (1.2–1.6)   | 1 class I study <sup>e26</sup>       |
| Eptinezumab                                                                                                   | 300 mg IV every 12 wk                                                                                                                                | 12 wk      | 1.56 (1.34–1.82) | 1 class I study <sup>e26</sup>       |
| Eptinezumab                                                                                                   | 300 mg IV every 12 wk                                                                                                                                | 24 wk      | 1.45 (1.27–1.67) | 1 class I study <sup>e26</sup>       |
| Erenumab                                                                                                      | 70 mg SC injection every 4 wk                                                                                                                        | 12 wk      | 1.71 (1.30–2.24) | 1 class I study <sup>e27</sup>       |
|                                                                                                               | 140 mg SC injection every 4 wk                                                                                                                       | 12 wk      | 1.75 (1.34–2.29) | 1 class I study <sup>e27</sup>       |
| Galcanezumab                                                                                                  | 240 mg loading dose followed by monthly 120 mg SC injection                                                                                          | 12 wk      | 2.30 (1.23–4.29) | 2 class I studies <sup>e21,e20</sup> |
|                                                                                                               | 240 mg SC injection per mo <sup>b</sup>                                                                                                              | 12 wk      | 1.79 (1.35–2.34) | 1 class I study <sup>e20</sup>       |
| OnabotulinumtoxinA                                                                                            | 155–195 units IM injections across corrugator, procerus, frontalis, temporalis, occipitalis, cervical paraspinal, and trapezius muscles, every 12 wk | 24 wk      | 1.31 (1.15–1.49) | 1 class I study <sup>e29</sup>       |

Abbreviation: RR = risk ratio; SC = subcutaneous.  
<sup>a</sup> Responder rate = A reduction in headache frequency of ≥50%.  
<sup>b</sup> Drug dosage is currently not available clinically or differs from the recommended dosages on the drug label.

**Episodic Migraine: Responder Rate (≥50% Reduction in Headache Frequency)**

*Moderate Confidence in the Evidence*

Galcanezumab (240 mg loading dose, followed by 120 mg monthly SC injections) is probably more effective than rimegepant (75 mg oral every other day) in reducing migraine frequency by ≥ 50% at 12 weeks (RR 1.21, 95% CI 1.11–1.31; 1 Class I study<sup>e33</sup>).

*Low Confidence in the Evidence*

Atorvastatin (oral, 40 mg/day) is possibly no more likely than valproate (oral, 500 mg/day)<sup>e54</sup> to reduce headache frequency by ≥ 50% (RR 0.90, 95% CI 0.67–1.23; 1 Class I study<sup>e55</sup>).

**Chronic Migraine: Change in Number of Headache Days per Month**

*Moderate Confidence in the Evidence*

Valproate (oral, 1,000 mg/day) is probably more likely than levetiracetam (oral, 1,000 mg/day) to reduce headache frequency at 12 weeks (RMD –7.00, 95% CI –10.30 to –3.8; 1 Class II study<sup>e54</sup>).

*Low Confidence in the Evidence*

Flunarizine (oral, 10 mg/day) is possibly more likely than topiramate (oral, 50 mg/day) to reduce headache frequency at 8 weeks (RMD –2.60, 95% CI –4.50 to –0.70; 1 Class II study<sup>e56</sup>).

Topiramate (oral, 100 mg/day) plus monthly lidocaine injection is possibly more effective than topiramate (oral, 100 mg/day) in reducing headache frequency at 12 weeks (SMD –0.71, 95% CI –1.17 to –0.27; 1 Class II study<sup>e57</sup>).

Topiramate (oral, 100 mg/day) plus methylprednisolone + lidocaine for 1 injection and monthly lidocaine for 2 injections is possibly more effective than topiramate (oral, 100 mg/day) in reducing headache frequency at 12 weeks (SMD –0.54, 95% CI –0.98 to –0.11; 1 Class II study<sup>e57</sup>).

**Chronic Migraine: Responder Rate (≥50% Reduction in Headache Frequency)**

*Low Confidence in the Evidence*

Topiramate (oral, 100 mg/day) plus monthly bilateral greater occipital nerve lidocaine injection (2%, 40 mg) is possibly more effective than topiramate (oral, 100 mg/day) alone in reducing headache frequency by ≥ 50% at 12 weeks (RR 1.62, 95% CI 1.04–2.56; 1 Class II study<sup>e57</sup>).

Topiramate (oral, 100 mg/day) plus bilateral greater occipital nerve injection with methylprednisolone (80 mg) + lidocaine (2%, 40 mg) for 1 injection and monthly lidocaine (2%, 40 mg) for 2 injections is possibly more effective than topiramate (oral, 100 mg/day) alone in reducing headache frequency by  $\geq 50\%$  at 12 weeks (RR 1.83, 95% CI 1.22–2.85; 1 Class II study<sup>e57</sup>).

### Episodic or Chronic Migraine: Change in Number of Headache Days

#### Moderate Confidence in the Evidence

Erenumab (70–140 mg SC injection monthly) is probably more effective than topiramate (oral, 50–100 mg/day) in reducing headache frequency at 24 weeks (SMD  $-0.44$ , 95% CI  $-0.58$  to  $-0.3$ ; 1 Class I study<sup>e58</sup>).

### Episodic or Chronic Migraine: Responder Rate ( $\geq 50\%$ Reduction in Headache Frequency)

#### Moderate Confidence in the Evidence

Erenumab (70–140 mg SC injection monthly) is probably more effective than topiramate (oral, 50–100 mg/day) in reducing headache frequency by  $\geq 50\%$  at 24 weeks (RR 1.78, 95% CI 1.5–2.11; 1 Class I study<sup>e58</sup>).

eAppendix 4 lists drugs for which there was insufficient evidence (very low confidence) to determine superiority to an active comparator for reducing headache frequency.

## Clinical Question 3

1. In adults (aged 18 years and older) with migraine, which pharmacologic preventive treatments, compared with placebo or other preventive pharmacologic treatments, improve patient-reported QOL on validated instruments?

### Episodic Migraine

For each drug, Table 6 presents the dose, time point, effect size, and evidence to support each conclusion about QOL outcomes when used for episodic migraine prevention vs placebo.

#### Active Comparator Studies

For episodic migraine, there is low confidence that zonisamide (oral, 200 mg/day) is possibly no more effective than topiramate (oral, 100 mg/day) in improving patient-reported QOL on the MIDAS at 12 weeks (RMD 1.10, 95% CI  $-1.59$  to  $3.79$ ; 1 Class II study<sup>e51</sup>). There is moderate confidence that galcanezumab (240 mg loading dose, 120 mg monthly SC injection) is probably more effective than rimegepant (75 mg oral every other day) in improving patient-reported QOL on the MSQ-RFR (RMD 5.2, 95% CI 1.9–8.6) and low confidence that it is possibly more likely to improve patient-reported QOL on the MSQ-RFP at 12 weeks (RMD 3.88, 95% CI 1.06–6.71; 1 Class I study<sup>e33</sup>). There is moderate confidence that

galcanezumab (240 mg loading dose, 120 mg monthly SC injection) is probably no more effective than rimegepant (75 mg oral every other day) in improving patient-reported QOL on the MSQ-EF (RMD 3.18, 95% CI  $-0.08$  to  $6.44$ ), and low confidence that it is possibly no more effective in improving patient-reported QOL on the MIDAS at 12 weeks (RMD  $-2.43$ , 95% CI  $-6.94$  to  $2.08$ ; 1 Class I study<sup>e33</sup>). eAppendix 4 lists drugs for which there was insufficient evidence to determine if the intervention was superior to placebo or another active treatment in improving patient-reported QOL.

### Chronic Migraine

For each drug, Table 7 presents the dose, time point, effect size, and evidence to support each conclusion about QOL outcomes when used for chronic migraine prevention vs placebo.

#### Active Comparator Studies

For chronic migraine, valproate (oral, 1,000 mg/day) is possibly more effective than levetiracetam (oral, 1,000 mg/day) in improving patient-reported QOL on the MIDAS at 12 weeks (RMD  $-15.50$ , 95% CI  $-23.80$  to  $-7.20$ ; low confidence, 1 Class II study<sup>e54</sup>). OnabotulinumtoxinA (200 units IM) is possibly no more effective than topiramate (oral, 100–200 mg/day) in improving patient-reported QOL on the HIT-6 at 12 weeks (RMD  $-0.43$ , 95% CI  $-1.35$  to  $0.5$ ; low confidence, 2 Class III studies).<sup>e69,e70</sup> There is low confidence that topiramate (oral, 100 mg/day) plus monthly lidocaine injections (40 mg, 2%) to bilateral greater occipital nerves are more effective than topiramate (oral, 100 mg/day) in improving patient-reported QOL on the HIT-6 at 12 weeks (RMD  $-3.7$ , 95% CI  $-6.34$  to  $-1.07$ ; 1 Class II study<sup>e57</sup>), and are possibly no more effective in improving patient-reported QOL on the MIDAS at 12 weeks (RMD  $-5.1$ , 95% CI  $-8.71$  to  $-1.49$ ; 1 Class II study<sup>e57</sup>). Topiramate (oral, 100 mg/day) plus methylprednisolone 80 mg and lidocaine (40 mg, 2%) injections to the bilateral greater occipital nerves are possibly more effective than topiramate (oral, 100 mg/day) in improving patient-reported QOL on the HIT-6 at 12 weeks (RMD  $-5.5$ , 95% CI  $-8.07$  to  $-2.94$ ) and the MIDAS at 12 weeks (RMD  $-4.8$ , 95% CI  $-8.41$  to  $-1.18$ , 1 Class II study<sup>e57</sup>). eAppendix 4 lists medications for which there was insufficient evidence to determine if the intervention was superior to placebo or another active treatment in improving patient-reported QOL.

### Episodic or Chronic Migraine

For each drug, eTable 3 presents the dose, time point, effect size, and evidence to support each conclusion about QOL outcomes when used for episodic or chronic migraine prevention.

There is low confidence that rimegepant (75 mg oral every other day) is possibly no more likely than placebo to improve patient-reported QOL on the MIDAS at 12 weeks (RMD

**Table 6** Episodic Migraine: Quality of Life Outcomes

| MIDAS                                                                                                                                   |                                                  |            |                           |                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------|---------------------------|-------------------------------------------------------|
| High confidence in the evidence (more effective than placebo in improving patient-reported quality of life on the MIDAS)                |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection | 12 wk      | −13.86 (−18.96 to −8.76)  | 2 class I studies <sup>20,21</sup>                    |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MIDAS)   |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Fremanezumab                                                                                                                            | 225 mg monthly SC injection                      | 12 wk      | −5.94 (−8.17 to −3.7)     | 2 class I studies <sup>36,37</sup>                    |
|                                                                                                                                         | 675 mg SC injection every 12 wk                  | 12 wk      | −5.29 (−7.54 to −3.04)    | 2 class I studies <sup>36,37</sup>                    |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection | 24 wk      | −8.04 (−10.06 to −6.02)   | 1 class I and 1 class II study <sup>22,e59</sup>      |
|                                                                                                                                         | 240 mg monthly SC injection <sup>a</sup>         | 24 wk      | −6.99 (−9.03 to −4.95)    | 1 class I and 1 class II study <sup>22,e59</sup>      |
| OnabotulinumtoxinA                                                                                                                      | 139 units IM injection across 6 muscle groups    | 12 wk      | −26.38 (−40.83 to −11.93) | 1 class II study <sup>e60</sup>                       |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MIDAS)        |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                       | 24 wk      | −2.1 (−3.30 to −0.90)     | 1 class II study <sup>e61</sup>                       |
|                                                                                                                                         | 140 mg monthly SC injection                      | 24 wk      | −2.8 (−4.00 to −1.70)     | 1 class II study <sup>e61</sup>                       |
| Oxcarbazepine                                                                                                                           | 1,200 mg oral per d                              | 15 wk      | −0.52 (−0.99 to −0.05)    | 1 class II study <sup>e11</sup>                       |
| HIT-6                                                                                                                                   |                                                  |            |                           |                                                       |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the HIT-6)   |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                       | 12 wk      | −1.87 (−2.56 to −1.18)    | 3 class I studies <sup>29,34,35</sup>                 |
|                                                                                                                                         |                                                  | 24 wk      | −2.10 (−2.85 to −1.36)    | 1 class I and 1 class II study <sup>e61,26</sup>      |
|                                                                                                                                         | 140 mg monthly SC injection                      | 12 wk      | −2.84 (−3.86 to −1.82)    | 2 class I studies <sup>29,e62</sup>                   |
|                                                                                                                                         |                                                  | 24 wk      | −2.18 (−2.93 to −1.44)    | 1 class I and 1 class II study <sup>e61,26</sup>      |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the HIT-6)        |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection | 12 wk      | −2.60 (−4.50 to −0.69)    | 1 class I and 1 class II study <sup>e63,e64</sup>     |
| MSQ-RFR                                                                                                                                 |                                                  |            |                           |                                                       |
| High confidence in the evidence (more effective than placebo in improving patient-reported quality of life on the MSQ-RFR)              |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection | 12 wk      | 8.82 (4.58–13.06)         | 2 class I and 1 class II studies <sup>20,21,e63</sup> |
|                                                                                                                                         | 240 mg loading dose, 120 mg monthly SC injection | 24 wk      | 8.18 (6.27–10.10)         | 2 class I studies <sup>22,23</sup>                    |
|                                                                                                                                         | 240 mg monthly SC injection <sup>a</sup>         | 24 wk      | 7.36 (5.42–9.29)          | 2 class I studies <sup>22,23</sup>                    |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MSQ-RFR) |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Atogepant                                                                                                                               | 10 mg oral once daily                            | 12 wk      | 9.9 (5.4–14.4)            | 1 class I study <sup>31</sup>                         |
| Atogepant                                                                                                                               | 30 mg oral once daily                            | 12 wk      | 10.2 (5.7–14.5)           | 1 class I study <sup>31</sup>                         |
| Atogepant                                                                                                                               | 60 mg oral once daily                            | 12 wk      | 10.8 (6.4–15.2)           | 1 class I study <sup>31</sup>                         |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MSQ-RFR)      |                                                  |            |                           |                                                       |
| Drug                                                                                                                                    | Dose, frequency, and route                       | Time point | RMD (95% CI)              | Evidence                                              |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                       | 12 wk      | 4.04 (0.50–7.59)          | 2 class I studies <sup>34,35</sup>                    |
|                                                                                                                                         |                                                  | 24 wk      | 5.10 (2.80–7.40)          | 1 class II study <sup>e61</sup>                       |
|                                                                                                                                         | 140 mg monthly SC injection                      | 24 wk      | 6.50 (4.20–8.80)          | 1 class II study <sup>e61</sup>                       |

Continued

**Table 6** Episodic Migraine: Quality of Life Outcomes (*continued*)

| MSQ-RFP                                                                                                                                  |                                                  |            |                        |                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------|------------------------|--------------------------------------------------|
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MSQ-RFP)  |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Galcanezumab                                                                                                                             | 240 mg loading dose, 120 mg monthly SC injection | 12 wk      | 6.07 (4.23–7.89)       | 1 class I and 1 class II study <sup>20,e63</sup> |
|                                                                                                                                          |                                                  | 24 wk      | 6.7 (4.54–8.86)        | 1 class II study <sup>e59</sup>                  |
|                                                                                                                                          | 240 mg monthly SC injection <sup>a</sup>         | 24 wk      | 5.72 (3.76–7.68)       | 1 class II study <sup>e59</sup>                  |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MSQ-RFP)       |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Erenumab                                                                                                                                 | 70 mg monthly SC injection                       | 12 wk      | 2.40 (0.55–5.36)       | 2 class I studies <sup>34,35</sup>               |
|                                                                                                                                          |                                                  | 24 wk      | 4.20 (2.20–6.30)       | 1 class II study <sup>e61</sup>                  |
|                                                                                                                                          | 140 mg monthly SC injection                      | 24 wk      | 5.40 (3.40–7.50)       | 1 class II study <sup>e61</sup>                  |
| MSQ-EF                                                                                                                                   |                                                  |            |                        |                                                  |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MSQ-EF)   |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Galcanezumab                                                                                                                             | 240 mg loading dose, 120 mg monthly SC injection | 24 wk      | 8.41 (6.40–10.42)      | 1 class II study <sup>e59</sup>                  |
|                                                                                                                                          | 240 mg monthly SC injection <sup>a</sup>         | 24 wk      | 7.53 (5.49–9.57)       | 1 class II study <sup>e59</sup>                  |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MSQ-EF)        |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Erenumab                                                                                                                                 | 70 mg monthly SC injection                       | 12 wk      | 3.73 (1.30–6.16)       | 2 class I studies <sup>34,35</sup>               |
|                                                                                                                                          |                                                  | 24 wk      | 5.20 (2.80–7.60)       | 1 class II study <sup>e61</sup>                  |
|                                                                                                                                          | 140 mg monthly SC injection                      | 24 wk      | 6.70 (4.40–9.10)       | 1 class II study <sup>e61</sup>                  |
| Galcanezumab                                                                                                                             | 240 mg loading dose, 120 mg monthly SC injection | 12 wk      | 5.91 (0.86–10.95)      | 1 class I and 1 class II study <sup>20,e63</sup> |
| MPFID-EA                                                                                                                                 |                                                  |            |                        |                                                  |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MPFID-EA) |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Erenumab                                                                                                                                 | 140 mg monthly SC injection                      | 12 wk      | –3.90 (–6.10 to –1.70) | 1 class I study <sup>28</sup>                    |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MPFID-EA)      |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Erenumab                                                                                                                                 | 70 mg monthly SC injection                       | 24 wk      | –2.20 (–3.30 to –1.20) | 1 class I study <sup>25</sup>                    |
|                                                                                                                                          | 140 mg monthly SC injection                      | 24 wk      | –2.60 (–3.60 to –1.50) | 1 class I study <sup>25</sup>                    |
| MPFID-PI                                                                                                                                 |                                                  |            |                        |                                                  |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MPFID-PI)      |                                                  |            |                        |                                                  |
| Drug                                                                                                                                     | Dose, frequency, and route                       | Time point | RMD (95% CI)           | Evidence                                         |
| Erenumab                                                                                                                                 | 70 mg monthly SC injection                       | 24 wk      | –1.90 (–3.00 to –0.80) | 1 class I study <sup>25</sup>                    |
|                                                                                                                                          |                                                  | 12 wk      | –3.5 (–5.70 to –1.20)  | 1 class I study <sup>28</sup>                    |
|                                                                                                                                          | 140 mg monthly SC injection                      | 24 wk      | –2.40 (–3.50 to –1.40) | 1 class I study <sup>25</sup>                    |

Abbreviation: HIT-6 = Headache Impact Test; MIDAS = Migraine Disability Assessment Scale; MPFID-EA = Migraine Physical Function Impact Diary–Everyday Activities; MPFID-PI = Migraine Physical Function Impact Diary–Physical Impairment; MSQ-EF = Migraine Specific Quality of Life Questionnaire–Emotional Function; MSQ-RFP = Migraine Specific Quality of Life Questionnaire–Role Function Preventive; MSQ-RFR = Migraine Specific Quality of Life Questionnaire–Role Function Restriction; RMD = raw mean difference; SC = subcutaneous.

<sup>a</sup> Drug dosage is currently not available clinically or differs from the recommended dosages on the drug label.

–0.01, 95% CI –4.7 to 4.5; 1 Class I study<sup>e71</sup>). There is moderate confidence that erenumab (70–140 mg monthly SC injection) is probably more effective than topiramate (oral,

50–100 mg/day) in improving patient-reported QOL on the HIT-6 at 24 weeks (RMD –3.2, 95% CI –4.3 to –2.1; 1 Class I study<sup>e58</sup>).



## Chronic Migraine With Medication Overuse

There is moderate confidence that acute medication withdrawal with onabotulinumtoxinA injection (155 units IM) is probably no more effective than acute medication withdrawal with onabotulinumtoxinA injection (17.5 units IM) in improving patient-reported QOL on the HIT-6 at 12 weeks (RMD 0, 95% CI -1.5 to 1.6; 1 Class I study<sup>e72</sup>).

## Strengths and Limitations

This SR has several notable strengths. It was conducted using a comprehensive and transparent search strategy, applied explicit eligibility criteria, and used standardized methods for study selection, data extraction, and risk-of-bias assessment. The inclusion of both episodic and chronic migraine populations enhances clinical relevance, and the use of structured evidence grading provides clarity regarding the certainty of effect estimates. By synthesizing data across multiple pharmacologic classes—including newer targeted therapies and established agents—the review offers a broad and contemporary overview of preventive treatment options. The emphasis on patient-centered outcomes, such as headache frequency and quality of life, further strengthens its applicability to clinical decision-making.

However, several limitations should be acknowledged. The quality and heterogeneity of included trials—particularly with respect to study design, duration, outcome definitions, and comparator selection—may limit the precision and comparability of pooled estimates. As standards for clinical trials and reporting have evolved over the past 50 years, contemporary trials are much more likely to be rated at lower ROB using our current standards. Many studies were industry-sponsored, introducing potential reporting bias. Short follow-up durations in randomized trials constrain conclusions about long-term efficacy and safety. In addition, underrepresentation of certain patient subgroups (e.g., those with significant comorbidities, older adults, or diverse racial and socioeconomic backgrounds) may limit generalizability. Publication bias and selective outcome reporting cannot be excluded. Resource limitations and the complexity of our process prevent us from including the most recently published RCTs in this analysis.

Finally, we must remember that the term “chronic migraine” was only incorporated in the 2nd edition of the International Classification of Headache Disorders in 2004.<sup>e73</sup> Chronic migraine was based on the concept of transformation from infrequent to more frequent migraine attacks described by Silberstein.<sup>e74</sup> The dichotomization of migraine into “chronic migraine” and nonchronic (i.e., “episodic migraine”) has therefore only been used in the most recent clinical trials evaluating preventive treatments. Many of the older trials were not conducted specifically in people with episodic or chronic migraine according to current definitions and include people with migraine of various frequencies. In addition, most

trials include participants with migraine with and without aura, without specifying the effect of treatment on attacks with vs without aura.

## Suggestions for Future Research

Even with recent advances in preventive treatments, many people with migraine still do not respond adequately to current treatment options. There are several promising new treatment targets that need to be explored, as several recent reviews have addressed.<sup>e75-e77</sup>

Methysergide, a 5-hydroxytryptamine antagonist, was introduced as a preventive drug for migraine in 1959, but use was limited by retroperitoneal fibrosis.<sup>e78</sup> In 1966, a case series on propranolol use in angina reported that 1 patient had relief of vascular headaches, and propranolol eventually became a standard therapy for migraine prevention.<sup>e79</sup> Several additional oral drugs became widely used over time. An important advance in pharmacologic migraine prevention occurred in 2010 with FDA approval of onabotulinumtoxinA for chronic migraine prevention. This was followed in 2018 by US approval of erenumab, a monoclonal antibody directed against the CGRP receptor. FDA approval of an oral CGRP antagonist for migraine prevention, atogepant, followed in 2021.

CGRP antagonists were a major advance in migraine prevention, but unfortunately not all with migraine respond well to them. There is still a large unmet treatment need, with the following being promising areas for future research:

1. Pituitary adenylate cyclase-activating polypeptides (PACAPs) may be important in migraine pathophysiology. PACAP-38 is found in human trigeminal neurons, can trigger headache in those with migraine, is elevated during migraine attacks, and is normalized by sumatriptan with attack resolution. Like CGRP, it is a vasodilator. Although a clinical trial in migraine with a PACAP antagonist was negative, further research is indicated.<sup>e80,e81</sup>
2. The hypothalamus may be involved in the early stages of migraine attacks. In animal models, suvorexant, a dual orexin receptor antagonist, has been shown to reduce the activity of second order neurons originating in the trigeminocervical complex, increase the threshold for cortical spreading depression, and reduce allodynia responses. A clinical trial with an orexin antagonist was negative, but more research is warranted.<sup>e75,e76</sup>
3. Human studies have shown a reduction in blood glutamate levels with some migraine preventives. The NMDA receptor (NMDA-R) may be particularly relevant to migraine, and 2 small, randomized placebo-controlled clinical trials with the NMDA-R blocker memantine have shown positive results.<sup>e75</sup> More research is needed.
4. More research is needed in the utility of medication combinations for migraine preventive therapy. There is evidence, for example, that a combination of

**Table 7** Chronic Migraine: Quality of Life Outcomes

| MIDAS                                                                                                                                   |                                                                                                                                                         |            |                          |                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------|--------------------------------------|
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MIDAS)   |                                                                                                                                                         |            |                          |                                      |
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                              | Time point | RMD (95% CI)             | Evidence                             |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                                                                                                                              | 12 wk      | -11.9 (-19.3 to -4.4)    | 1 class I study <sup>e65</sup>       |
|                                                                                                                                         | 140 mg monthly SC injection                                                                                                                             | 12 wk      | -12.2 (-19.7 to -4.8)    | 1 class I study <sup>e65</sup>       |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MIDAS)        |                                                                                                                                                         |            |                          |                                      |
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                              | Time point | RMD (95% CI)             | Evidence                             |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection                                                                                                        | 12 wk      | -11.54 (-20.32 to -2.76) | 2 class I studies <sup>e21,e20</sup> |
| Topiramate                                                                                                                              | 50–200 mg oral per d                                                                                                                                    | 16 wk      | -29.0 (-55.3 to -2.7)    | 1 class II study <sup>e23</sup>      |
| HIT-6                                                                                                                                   |                                                                                                                                                         |            |                          |                                      |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the HIT-6)   |                                                                                                                                                         |            |                          |                                      |
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                              | Time point | RMD (95% CI)             | Evidence                             |
| Eptinezumab                                                                                                                             | 300 mg IV every 12 wk                                                                                                                                   | 12 wk      | -2.9 (-3.9 to -1.8)      | 1 class I study <sup>e26</sup>       |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                                                                                                                              | 12 wk      | -2.5 (-3.7 to -1.2)      | 1 class I study <sup>e65</sup>       |
| Erenumab                                                                                                                                | 140 mg monthly SC injection                                                                                                                             | 12 wk      | -2.5 (-3.7 to -1.2)      | 1 class I study <sup>e65</sup>       |
| Fremanezumab                                                                                                                            | 675 mg SC injection at baseline, 225 mg at 4 wk, 225 mg at 8 wk                                                                                         | 12 wk      | -2.13 (-2.93 to -1.33)   | 2 class I studies <sup>e17,e18</sup> |
| OnabotulinumtoxinA                                                                                                                      | 155–195 units<br>IM injections across corrugator, procerus, frontalis, temporalis, occipitalis, cervical paraspinal, and trapezius muscles, every 12 wk | 24 wk      | -2.40 (-3.09 to -1.70)   | 2 class I studies <sup>e21,e22</sup> |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the HIT-6)        |                                                                                                                                                         |            |                          |                                      |
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                              | Time point | RMD (95% CI)             | Evidence                             |
| Eptinezumab                                                                                                                             | 100 mg IV every 12 wk                                                                                                                                   | 12 wk      | -1.7 (-2.8 to -0.7)      | 1 class I study <sup>e26</sup>       |
| Fremanezumab                                                                                                                            | 675 mg SC injection every 12 wk                                                                                                                         | 12 wk      | -1.77 (-2.57 to -0.97)   | 2 class I studies <sup>e18,e66</sup> |
| MSQ-RFR                                                                                                                                 |                                                                                                                                                         |            |                          |                                      |
| High confidence in the evidence (more effective than placebo in improving patient-reported quality of life on the MSQ-RFR)              |                                                                                                                                                         |            |                          |                                      |
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                              | Time point | RMD (95% CI)             | Evidence                             |
| OnabotulinumtoxinA                                                                                                                      | 155–195 units<br>IM injections across corrugator, procerus, frontalis, temporalis, occipitalis, cervical paraspinal, and trapezius muscles every 12 wk  | 24 wk      | 8.40 (6.01–10.76)        | 2 class I studies <sup>e67</sup>     |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MSQ-RFR) |                                                                                                                                                         |            |                          |                                      |
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                              | Time point | RMD (95% CI)             | Evidence                             |
| Atogepant                                                                                                                               | 30 mg oral twice daily <sup>a</sup>                                                                                                                     | 12 wk      | 8 (4.3–11.6)             | 1 class I study <sup>e25</sup>       |
| Atogepant                                                                                                                               | 60 mg oral once daily                                                                                                                                   | 12 wk      | 6.1 (2.5–9.8)            | 1 class I study <sup>e25</sup>       |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                                                                                                                              | 12 wk      | 6 (2.3–9.6)              | 1 class I study <sup>e65</sup>       |
|                                                                                                                                         | 140 mg monthly SC injection                                                                                                                             | 12 wk      | 7.4 (3.7–11)             | 1 class I study <sup>e65</sup>       |
| Fremanezumab                                                                                                                            | 225 mg monthly SC injection                                                                                                                             | 12 wk      | 6.3 (3.13–9.47)          | 1 class I study <sup>e68</sup>       |
|                                                                                                                                         | 675 mg SC injection every 12 wk                                                                                                                         | 12 wk      | 5.6 (2.78–8.42)          | 1 class I study <sup>e68</sup>       |
| Galcanezumab                                                                                                                            | 240 mg monthly SC injection <sup>a</sup>                                                                                                                | 12 wk      | 6.30 (3.00–9.60)         | 1 class I study <sup>e20</sup>       |

Continued

**Table 7** Chronic Migraine: Quality of Life Outcomes (continued)

| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MSQ-RFR)      |                                                                                                                                                        |            |                   |                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------|--------------------------------------|
| Drug                                                                                                                                    | Dose, frequency, and route                                                                                                                             | Time point | RMD (95% CI)      | Evidence                             |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection                                                                                                       | 12 wk      | 9.26 (0.65–17.87) | 2 class I studies <sup>e21,e20</sup> |
| MSQ-RFP                                                                                                                                 |                                                                                                                                                        |            |                   |                                      |
| High confidence in the evidence (more effective than placebo in improving patient-reported quality of life on the MSQ-RFP)              |                                                                                                                                                        |            |                   |                                      |
| OnabotulinumtoxinA                                                                                                                      | 155–195 units<br>IM injections across corrugator, procerus, frontalis, temporalis, occipitalis, cervical paraspinal, and trapezius muscles every 12 wk | 24 wk      | 6.70 (4.35–9.01)  | 2 class I studies <sup>e67</sup>     |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MSQ-RFP) |                                                                                                                                                        |            |                   |                                      |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection                                                                                                       | 12 wk      | 7.00 (4.20–9.80)  | 1 class I study <sup>e20</sup>       |
|                                                                                                                                         | 240 mg monthly SC injection <sup>a</sup>                                                                                                               | 12 wk      | 5.10 (2.30–7.90)  | 1 class I study <sup>e20</sup>       |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MSQ-RFP)      |                                                                                                                                                        |            |                   |                                      |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                                                                                                                             | 12 wk      | 4.1 (0.9–7.4)     | 1 class I study <sup>e65</sup>       |
|                                                                                                                                         | 140 mg monthly SC injection                                                                                                                            | 12 wk      | 4.9 (1.7–8.2)     | 1 class I study <sup>e65</sup>       |
| Fremanezumab                                                                                                                            | 225 mg monthly SC injection                                                                                                                            | 12 wk      | 3.9 (1.46–6.34)   | 1 class I study <sup>e68</sup>       |
|                                                                                                                                         | 675 mg SC injection every 12 wk                                                                                                                        | 12 wk      | 4.3 (1.81–6.79)   | 1 class I study <sup>e68</sup>       |
| MSQ-EF                                                                                                                                  |                                                                                                                                                        |            |                   |                                      |
| High confidence in the evidence (more effective than placebo in improving patient-reported quality of life on the MSQ-EF)               |                                                                                                                                                        |            |                   |                                      |
| OnabotulinumtoxinA                                                                                                                      | 155–195 units<br>IM injections across corrugator, procerus, frontalis, temporalis, occipitalis, cervical paraspinal, and trapezius muscles every 12 wk | 24 wk      | 8.4 (5.56–11.37)  | 2 class I studies <sup>e67</sup>     |
| Moderate confidence in the evidence (probably more effective than placebo in improving patient-reported quality of life on the MSQ-EF)  |                                                                                                                                                        |            |                   |                                      |
| Erenumab                                                                                                                                | 70 mg monthly SC injection                                                                                                                             | 12 wk      | 8.3 (4.3–12.4)    | 1 class I study <sup>e65</sup>       |
|                                                                                                                                         | 140 mg monthly SC injection                                                                                                                            | 12 wk      | 8.9 (4.9–13)      | 1 class I study <sup>e65</sup>       |
| Low confidence in the evidence (possibly more effective than placebo in improving patient-reported quality of life on the MSQ-EF)       |                                                                                                                                                        |            |                   |                                      |
| Fremanezumab                                                                                                                            | 225 mg monthly SC injection                                                                                                                            | 12 wk      | 3.3 (0.24–6.36)   | 1 class I study <sup>e68</sup>       |
|                                                                                                                                         | 675 mg SC injection every 12 wk                                                                                                                        | 12 wk      | 3.9 (0.84–6.96)   | 1 class I study <sup>e68</sup>       |
| Galcanezumab                                                                                                                            | 240 mg loading dose, 120 mg monthly SC injection                                                                                                       | 12 wk      | 7.00 (3.20–10.80) | 1 class I study <sup>e20</sup>       |
|                                                                                                                                         | 240 mg monthly SC injection <sup>a</sup>                                                                                                               | 12 wk      | 6.60 (2.80–10.40) | 1 class I study <sup>e20</sup>       |

Abbreviations: HIT-6 = Headache Impact Test; MIDAS = Migraine Disability Assessment Scale; MSQ-EF = Migraine Specific Quality of Life Questionnaire–Emotional Function; MSQ-RFP = Migraine Specific Quality of Life Questionnaire–Role Function Preventive; MSQ-RFR = Migraine Specific Quality of Life Questionnaire–Role Function Restriction; RMD = raw mean difference; SC = subcutaneous.  
<sup>a</sup> Drug dosage is currently not available clinically or differs from the recommended dosages on the drug label.

- onabotulinumtoxinA and a CGRP antagonist is more beneficial for some patients that either alone.<sup>e82,e83</sup>
5. Polygenic migraine disorders might be ideal candidates for precision medicine, defined as care designed to optimize therapeutic benefit for groups of patients by using genetic or molecular profiling.<sup>e84</sup> If biomarkers are developed to identify patients who are likely to respond to a particular treatment, true precision medicine for those with migraine will become possible.

6. Prospective registry-based studies that evaluate for potential harms of exposure to migraine preventives in utero and through excretion in breastmilk are needed in pregnancy and lactation.

7. There is limited evidence on the cardiovascular safety of CGRP therapies, including monoclonal antibodies and gepants, in people with established vascular disease or recent vascular events. While data suggest a favorable cardiovascular safety profile in people without

cardiovascular risk factors, additional data are needed to establish their safety in people with cardiovascular disease.<sup>e85</sup>

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